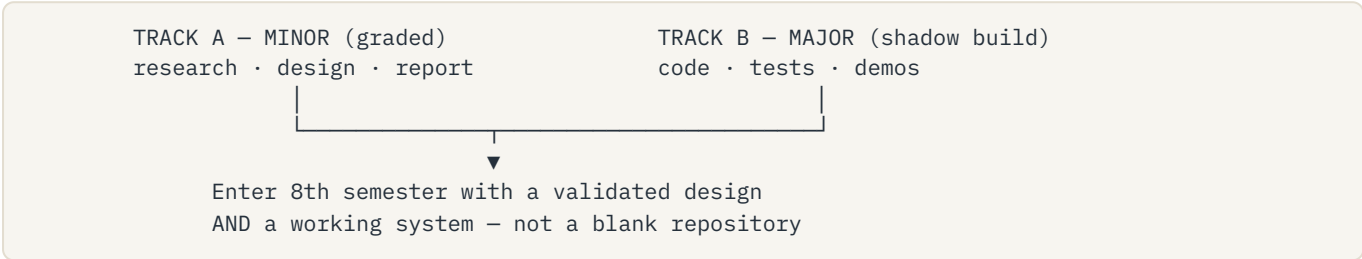


Roadmap — How the Project Gets From Here to Done

Project MAESTRO · Group 298 · as of 19 August 2026 (Week 5 of the Minor)

1. The Two-Track Model

Every week runs two tracks in parallel:



The one rule: if a week is compressed, **Track B slips, Track A does not**. Track B slipping costs comfort next semester; Track A slipping costs marks that cannot be recovered.

2. Where We Are

MINOR (7th sem, 12 weeks)		
W1 Title + synopsis	✓ done (WPR-1)	
W2 Setup + lit review begins	✓ done (WPR-2)	B: repo, env, local LLM JSON ✓
W3 Literature review I	✓ done	B: Action IR + registry ✓
W4 Lit review II + gap	✓ done (WPR-3/4)	B: 7 file verbs + dry-run ✓
W5 Requirements engineering	← YOU ARE HERE	B: safety engine v0 ✓ (early)
W6 Action IR + risk spec, SCOPE FREEZE		B: planner round trip ✓ (early)
W7 Architecture design		B: orchestrator + memory ✓ (early)
W8 Diagrams + design doc		B: voice I/O + browser verbs ⌚
W9 Evaluation methodology		B: eval harness, first 20 tasks
W10 Seed dataset (300+ pairs, κ)		B: dataset tooling + retrieval
W11 Report writing (Track B FROZEN)		
W12 Polish + viva		

Track B is running roughly **2–3 weeks ahead** of its own schedule (v0.2 with the LLM planner was a Week-6/7 deliverable; it exists now with 40 passing tests). That buffer is the insurance policy for Weeks 8–10, when the eval harness and dataset tooling compete with heavy report-writing weeks.

3. Minor — Remaining Weeks in Detail

WEEK	TRACK A (GRADED) DELIVERABLE	TRACK B (BUILD) DELIVERABLE
W5 (now)	Requirements chapter with traceability IDs; 8–10 full-form use cases; use-case diagram; feasibility study	(already delivered early: safety engine, consent gate, audit chain)
W6	Action IR v1.0 spec · risk taxonomy spec · trust model spec · SCOPE FREEZE — additions after this week go to Future Work only	(already delivered early: NL→execution round trip) — record the 60-second demo capture and show the guide

WEEK	TRACK A (GRADED) DELIVERABLE	TRACK B (BUILD) DELIVERABLE
W7	Component specs L0–L6; ER model; sequence flows; technology-justification chapter	<code>search.*</code> verbs; harden orchestrator edge cases
W8	All 10 graded diagrams (C4 ×3, DFD L0–L1, 3 sequence, state machine, ER, deployment) + algorithm/pseudocode chapter	Voice I/O (faster-whisper in, Piper out); <code>browser.open/extract/download</code> via Playwright
W9	Evaluation methodology chapter: 100-task suite design, formal metrics, baselines B0–B5, ablation matrix, user-study protocol (+ ethics answer)	Eval harness runs first 20 tasks → first real numbers for the report
W10	Dataset spec + annotation guidelines; 300+ seed pairs, 50 adversarial ; Cohen's κ on 50-pair overlap	Dataset validator, template expander, dedup; ChromaDB exemplar retrieval
W11	Report assembly only. Ch. 1–9 from material already written; plagiarism check; guide review with time to act	Freeze. Bug fixes + demo video only
W12	Final formatting; presentation; viva rehearsal (offline demo ×3 on the actual machine); WPR-12	—

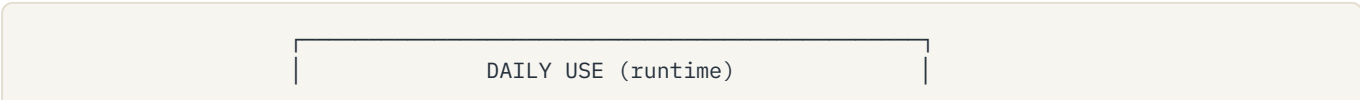
Minor exit criteria: 25+ papers with comparison matrix · frozen IR/risk/trust specs · full architecture + 10 diagrams · fixed evaluation methodology · 350 validated pairs · working vertical-slice demo · report + 12 WPRs.

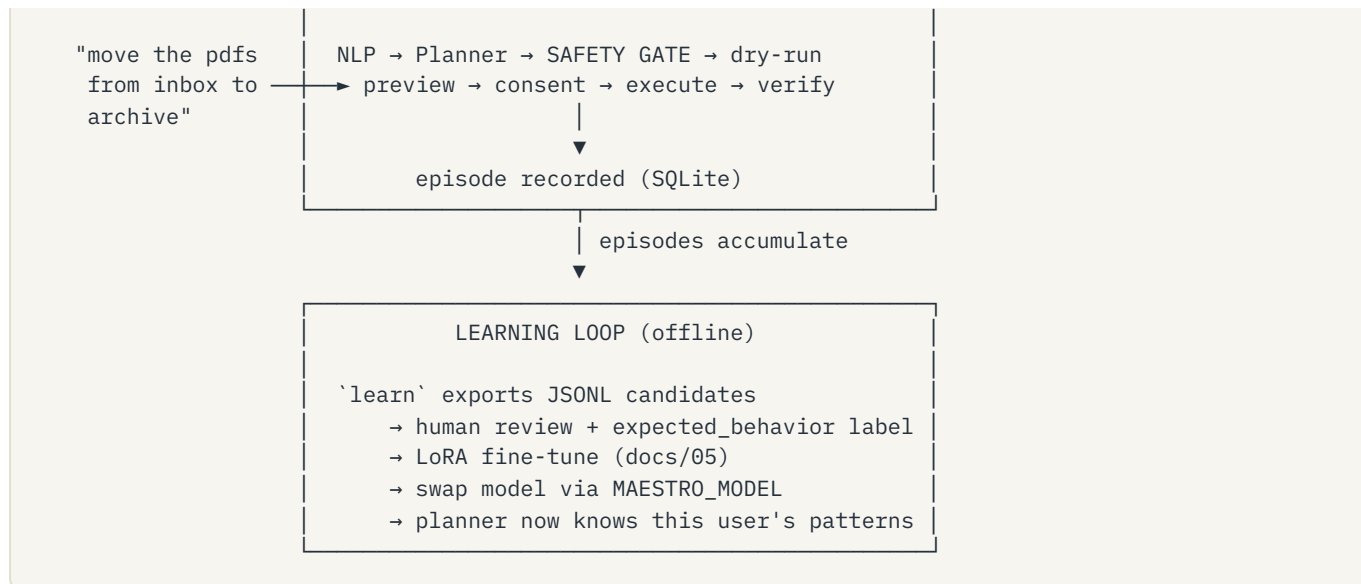
4. Major — 16 Weeks (8th Semester)

M1–M2	PLATFORM	Windows executor backend (win32/) for all P0 verbs; app/system/draft executors both OSes; differential test green
M3–M4	ROBUSTNESS	Preference learning, workflow retrieval, "the folder I used last time"; re-planning, budget guard, abort/rollback (M4 = designated buffer week)
M5–M6	DATASET	350 → 1,500 → 3,200 pairs (template + LLM-generated, every pair human-verified); 200 adversarial; splits frozen; public release prepared
M7–M8	FINE-TUNE	LoRA on Qwen2.5-3B (MLX local; Kaggle/Colab for sweeps); v2 + hyperparameter sweep + base-model ablations
M9–M10	BENCHMARK	100 tasks × B0–B4 × macOS, then Windows; cross-platform equivalence; ablations A0–A8
M11	ADVERSARIAL	40-case injection suite; per-control attribution table; **UER must be 0**; hard-block verification
M12	USER STUDY	$n \geq 15$; SUS + trust + NASA-TLX; interviews; stats
M13	GUI	Electron shell: plan preview, live progress, audit viewer, undo button (M13 = second buffer week)
M14	ANALYSIS	Consolidated results, significance tests, failure taxonomy
M15	WRITING	Major report; IEEE paper draft; dataset + code release
M16	SUBMISSION	Final report, demo video, viva, exhibition

Cut order if more than two weeks slip: (1) Electron GUI → keep TUI. (2) Windows P1 verbs → macOS-only for app/system, documented as limitation. (3) User study $n\ 20 \rightarrow 12$. **Never cut the adversarial safety evaluation or the fine-tune — they are the contribution.**

5. How the Finished System Works (the operating loop)





The runtime loop is the **product**; the learning loop is the **research experiment** (RQ4: can the fine-tuned local 3B match a frontier planner?). The two share one artifact — the episode store — which is why episodes were built into v0.2 now rather than bolted on later.

6. Milestones & Checkpoints

DATE (APPROX.)	MILESTONE	PROOF
✓ Aug 2026, W4	Gap analysis + novelty finalized	WPR-4, gap chapter
✓ Aug 2026, W5	Safety core coded & tested	40/40 tests, <code>maestro.cli</code> demo
Sep W6	Scope freeze ; specs v1.0; demo shown to guide	Signed-off spec docs + screen capture
Sep W8	Voice-driven demo; 10 diagrams done	Speak → gate → hear result
Sep W9	First real benchmark numbers (20-task pilot)	Metrics CSV
Oct W10	350 dataset pairs + κ reported	Dataset repo
Oct W11–12	Minor report + viva	Submission
8th sem M6	DeskPlan 3,200 frozen	HF/GitHub release candidate
8th sem M8	Fine-tune complete	M0–M5 comparison table
8th sem M11	UER = 0 on adversarial suite	Per-control attribution table
8th sem M16	Final submission	Report, paper, video, dataset

7. Standing Risks to Watch

RISK	MITIGATION ALREADY IN PLACE
Cross-platform doubles the work	Action IR keeps ~80% OS-independent; only L1 executors written twice; CI grep forbids <code>platform</code> above L1
Scope creep after W6	Scope freeze + Future Work file; say it out loud as a team
Free cloud tiers vanish	Local models cover every requirement; only the frontier <i>baseline</i> is lost

RISK	MITIGATION ALREADY IN PLACE
Ethics clearance blocks the M12 user study	Question raised with the guide now (Week 5) — long lead time
Refusal episodes poisoning the training set	<code>expected_behavior</code> labeling is mandatory before any training run (documented known issue)
Week-11 report crunch	"Write while reading" discipline since W3; report chapters already exist as drafts